# FINAL EXPLANATION: BIOLOGY GRADE 10

*Student Assessment Document*

| FINAL EXPLANATION: BIOLOGY \| GRADE 10 \| 2.1 PLANT NUTRITION | |
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| **Student Assessment Document** | |
| **Student Name** | ____________________________________ |
| **Class** | ____________________________________ |
| **Date** | ____________________________________ |

| INSTRUCTIONS FOR STUDENTS |
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| This is your Final Explanation document. Your task is to write a thorough, well-organised answer to the driving question using everything you have learned across the 10 lessons in this unit. Read the driving question carefully, then use each section below as a guide. For every claim you make, support it with evidence from your observations, experiments, or readings. A strong Final Explanation connects ideas across sections — show how wilting, photosynthesis, chlorophyll, and mineral nutrition are all linked in the life of your sukuma wiki plant. Write in full sentences and use correct scientific vocabulary throughout. |

| Section 1: Why Does the Sukuma Wiki Wilt at Midday — Even When the Soil Is Wet? | |
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| **Explain what is happening inside the sukuma wiki leaves during the hottest part of the day. In your answer:**<br>– **Describe what transpiration is and where it occurs in the leaf.**<br>– **Explain why transpiration rates are highest at midday (think about temperature, sunlight intensity, and stomata).**<br>– **Explain why wilting occurs even when the soil is moist — what does this tell us about the RATE of water loss versus water uptake?**<br>– **Use evidence from your investigations (e.g., the Vaseline-leaf activity or the** | During the midday heat in Nairobi, the sukuma wiki wilts because it is losing water through transpiration faster than its roots can absorb water from the soil. Transpiration is the evaporation of water vapour from the leaves, mainly through tiny pores called stomata on the lower surface of the leaf. At midday, high temperatures increase the kinetic energy of water molecules inside the leaf, causing them to evaporate rapidly. At the same time, bright sunlight causes the stomata to open wide to allow carbon dioxide in for photosynthesis. With stomata wide open and temperatures high, the rate of transpiration peaks. Even though the soil is moist, the roots cannot absorb and deliver water up the stem quickly enough to replace what is being lost. This creates a water deficit in the leaf cells — the cells lose turgor pressure and become flaccid, causing the leaf to wilt. In our Vaseline experiment, leaves coated on the underside (where most stomata are) wilted much more slowly than untreated leaves, confirming that stomata are the main exit point for water vapour. Overnight, temperatures drop, sunlight disappears, and the stomata close. Transpiration almost stops, allowing the roots to slowly reabsorb water from the moist soil, refilling the leaf cells and restoring turgor pressure. That is why the plant looks healthy again each morning — the damage is reversible because the soil still contains water. |

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| **potometer data) to support your explanation.**<br>**– Explain why the plant recovers overnight — what changes between midday and the following morning?** | |

## Section 2: What Is Photosynthesis, and Why Does the Plant Need Sunlight?

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| **Explain the process of photosynthesis in clear detail. In your answer:**<br>**– State the raw materials (inputs) and the products (outputs) of photosynthesis.**<br>**– Write the word equation and, if you can, the balanced chemical equation.**<br>**– Explain the role of sunlight — why can't the plant make food without it?**<br>**– Explain where in the cell photosynthesis takes place and why.**<br>**– Use your observations from the variegated leaf starch test or the light-intensity investigation to support your explanation.** | Photosynthesis is the process by which green plants use light energy to convert carbon dioxide and water into glucose and oxygen. The word equation is: Carbon dioxide + Water → Glucose + Oxygen. The balanced chemical equation is: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. Sunlight provides the energy needed to break apart water molecules and to power the chemical reactions that build glucose from carbon dioxide. Without light, these reactions cannot proceed — the plant has no energy source to drive them. Photosynthesis takes place in the chloroplasts, which are found mainly in the mesophyll cells of the leaf. Chloroplasts contain a green pigment called chlorophyll, which absorbs light energy — particularly red and blue wavelengths — and uses it to drive photosynthesis. In our variegated leaf experiment, only the green parts of the leaf (which contained chlorophyll) turned blue-black when tested with iodine solution, proving that starch was only produced where chlorophyll was present and light was absorbed. In our light-intensity investigation, we found that as we moved the lamp closer to the Elodea water plant — increasing light intensity — the rate of oxygen bubble production increased, showing that more photosynthesis was occurring. This directly explains why the sukuma wiki under the mango tree grows more slowly: the shade reduces light intensity, so the plant photosynthesises less, produces less glucose, and therefore has less energy and materials for growth. |

## Section 3: Why Do the Leaves Turn Pale Yellow-Green in the Shade?

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| **Explain why the sukuma wiki leaves gradually become pale yellowish-green when the plant is moved under the mango tree. In your answer:**<br>**– Explain what chlorophyll is and what it does.**<br>**– Explain how light affects chlorophyll production in** | The green colour of sukuma wiki leaves comes from chlorophyll, a pigment found inside chloroplasts that absorbs light energy to power photosynthesis. When the plant is moved into heavy shade under the mango tree, it receives very little light. Over time, the plant produces less chlorophyll — possibly because chlorophyll synthesis itself requires light, or because the plant breaks down existing chlorophyll faster than it replaces it when light is scarce. With less chlorophyll, the leaves appear pale yellowish-green or even yellow, a condition sometimes called etiolation or chlorosis. Less chlorophyll means less light energy is captured, so the rate of photosynthesis falls even further. With less glucose being made, the plant has less energy for growth and less raw material for building new cells — which is why the shaded sukuma wiki also grows more slowly. However, yellowing can also be caused by a deficiency of certain mineral ions, particularly magnesium. Magnesium ions ($Mg^{2+}$) are an essential component of the chlorophyll molecule itself. In our mineral deficiency investigation, plants grown in nutrient solution lacking magnesium developed yellowing |

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| **the leaf.**<br>– **Connect the loss of chlorophyll to the reduced rate of photosynthesis and slower growth.**<br>– **Mention any other factors (such as mineral nutrients) that could also cause yellowing, and explain the evidence for this from your investigations.** | leaves (chlorosis) similar to what we observed in the shaded plant, even though they had plenty of light. This reminds us that a plant needs both light AND the right mineral nutrients to stay green and photosynthesise efficiently. |

| Section 4: What Does the Plant Need From the Soil and Air to Grow? | |
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| **Explain the full range of inputs a sukuma wiki plant needs — not just for photosynthesis, but for healthy growth overall. In your answer:**<br>– **Identify the raw materials absorbed from the soil (water and mineral ions) and from the air (carbon dioxide).**<br>– **Explain why mineral ions such as nitrates, magnesium, and phosphates are essential, giving a specific role for each.**<br>– **Connect mineral nutrition to the process of photosynthesis and to plant health.**<br>– **Use evidence from your hydroponics or mineral deficiency experiment to support your claims.** | For healthy growth, the sukuma wiki plant needs water and dissolved mineral ions from the soil, and carbon dioxide from the air. Water is absorbed by root hair cells through osmosis and is used as a raw material in photosynthesis. It also keeps cells turgid, as we saw in Section 1. Carbon dioxide enters the leaf through the stomata and provides the carbon atoms that are built into glucose during photosynthesis. Mineral ions are absorbed from the soil water by active transport in the root hair cells. Three key mineral ions are: (1) Nitrate ions ($NO_3^-$) — used to make amino acids and proteins, which are needed for cell growth and for building enzymes including those used in photosynthesis. Plants without nitrates show stunted growth and yellowing of older leaves. (2) Magnesium ions ($Mg^{2+}$) — a core part of the chlorophyll molecule. Without magnesium, the plant cannot make enough chlorophyll and leaves turn yellow (chlorosis), directly reducing the rate of photosynthesis. (3) Phosphate ions ($PO_4^{3-}$) — needed for making ATP (the energy currency of the cell) and for DNA. Without phosphates, plants show poor root development and purple-tinged leaves. In our hydroponics experiment, sukuma wiki seedlings grown in complete nutrient solution remained healthy and dark green, while those grown in solutions lacking nitrates or magnesium showed clear symptoms of deficiency within two weeks. This demonstrates that even if light and water are plentiful, the plant cannot grow properly without the right mineral ions from the soil. |

| Section 5: Pulling It All Together — Answering the Driving Question | |
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| **Now bring all your ideas together to give a complete** | At the start of this unit, the wilting sukuma wiki was puzzling because the soil was wet — so why was the plant drooping? Now we can fully explain it. The sukuma wiki wilts at midday because the high temperature and intense sunlight cause the stomata to open |

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| **answer to the driving question. Your answer should:**<br>– **Explain the wilting phenomenon and why it is reversible.**<br>– **Explain why shade solves the wilting problem but creates new problems (pallor, slow growth).**<br>– **Explain what a plant truly needs — light, water, carbon dioxide, and mineral ions — and how these inputs connect to photosynthesis and healthy growth.**<br>– **Reflect on what surprised you or what you now understand that you did not understand at the start of the unit.** | wide and transpiration to surge. Water is lost from the leaves faster than the roots can absorb it from the soil, so leaf cells lose turgor and the plant wilts. The moist soil is not the problem — the RATE of water movement is. Overnight, when temperatures drop and stomata close, the plant reabsorbs water and recovers — proving the process is reversible and the plant is not permanently damaged. Moving the plant into the shade of the mango tree reduces transpiration and stops wilting, but it introduces a new problem: far less light reaches the leaves. With less light, the chloroplasts absorb less energy, the rate of photosynthesis falls, and the plant produces less glucose. Less glucose means less energy for growth and fewer building blocks for new cells, so the plant grows slowly. Over weeks, the plant also produces less chlorophyll, causing the leaves to turn pale yellowish-green. A truly healthy sukuma wiki therefore needs a careful balance: enough sunlight for photosynthesis and chlorophyll production, enough water to keep cells turgid and supply raw materials, enough carbon dioxide from the air to build glucose, and enough mineral ions — especially nitrates for proteins and magnesium for chlorophyll — absorbed from the soil. Photosynthesis is the central process that links all of these needs: it captures light energy and uses it to turn carbon dioxide and water into glucose, which the plant uses for energy (respiration) and as a building material for growth. What surprised me most was that wilting is not simply about drought — it is about the rate of water loss versus uptake. It also surprised me that shade, which feels gentler, can be just as damaging in the long run as scorching sun, because light is the energy source for all plant food-making. The sukuma wiki in the kitchen garden in Nairobi taught us that plants are constantly balancing their need for light, water, and nutrients — and that understanding these needs helps us grow healthier crops and understand life itself. |

## RUBRIC

| Criterion | Excellent (4) | Proficient (3) | Developing (1–2) |
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| Scientific Accuracy of Photosynthesis | Correctly states all inputs ($CO_2$, $H_2O$, light) and outputs (glucose, $O_2$); writes the correct word and chemical equation; accurately explains the role of chlorophyll and chloroplasts; links photosynthesis to energy for growth with no misconceptions. | Correctly identifies most inputs and outputs; writes the word equation correctly; explains the role of light and chlorophyll with minor omissions; mostly links photosynthesis to plant growth. | Identifies some inputs or outputs but makes errors (e.g., omits water or confuses respiration with photosynthesis); equation is incomplete or incorrect; limited connection to growth. |
| Explanation of the Wilting Phenomenon | Clearly and accurately explains transpiration, stomatal opening, and the imbalance between water loss and uptake; explains why moist soil does not prevent wilting; explains overnight recovery using scientific reasoning about turgor pressure. | Explains transpiration and links it to wilting; mentions stomata and temperature; explains recovery but may not explicitly discuss turgor pressure or rate imbalance. | States that water is lost but does not explain the mechanism; may incorrectly claim the soil is dry or that wilting is permanent damage; recovery is not explained or is explained incorrectly. |
| Use of Evidence from Investigations | Integrates specific evidence from at least three investigations (e.g., Vaseline-leaf activity, starch test, light-intensity experiment, mineral | References evidence from at least two investigations; evidence is relevant and mostly supports the claims made, though the connection may not always be | Mentions one investigation or refers to evidence vaguely (e.g., 'in our experiment'); evidence is not clearly linked to the scientific claim being made. |

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| | deficiency/hydroponics); evidence is used to directly support claims throughout all sections. | explicit. | |
| Mineral Nutrition and Plant Health | Accurately identifies at least three mineral ions (nitrates, magnesium, phosphates), states a specific role for each, and connects mineral deficiency to observable symptoms (e.g., chlorosis, stunted growth); links magnesium specifically to chlorophyll and photosynthesis. | Identifies two mineral ions with correct roles; connects at least one deficiency to an observable symptom; partially links minerals to photosynthesis or growth. | Names one or two minerals without explaining their specific roles; symptoms of deficiency are missing or incorrect; no clear link to photosynthesis. |
| Synthesis and Use of Scientific Vocabulary | Integrates all major concepts (transpiration, stomata, turgor, photosynthesis, chlorophyll, chloroplasts, mineral ions, glucose) into a coherent, well-structured narrative that fully answers the driving question; uses all key vocabulary correctly and consistently. | Addresses most major concepts and uses most key vocabulary correctly; the overall answer is coherent but one or two concepts are underdeveloped or a term is used slightly inaccurately. | Addresses fewer than half the major concepts; key vocabulary is missing, misused, or used inconsistently; the answer does not fully address the driving question. |